MACHINE LEARNING & CAUSALITY

The Impact of Financial Crises on Growth

— and where the field has gone since

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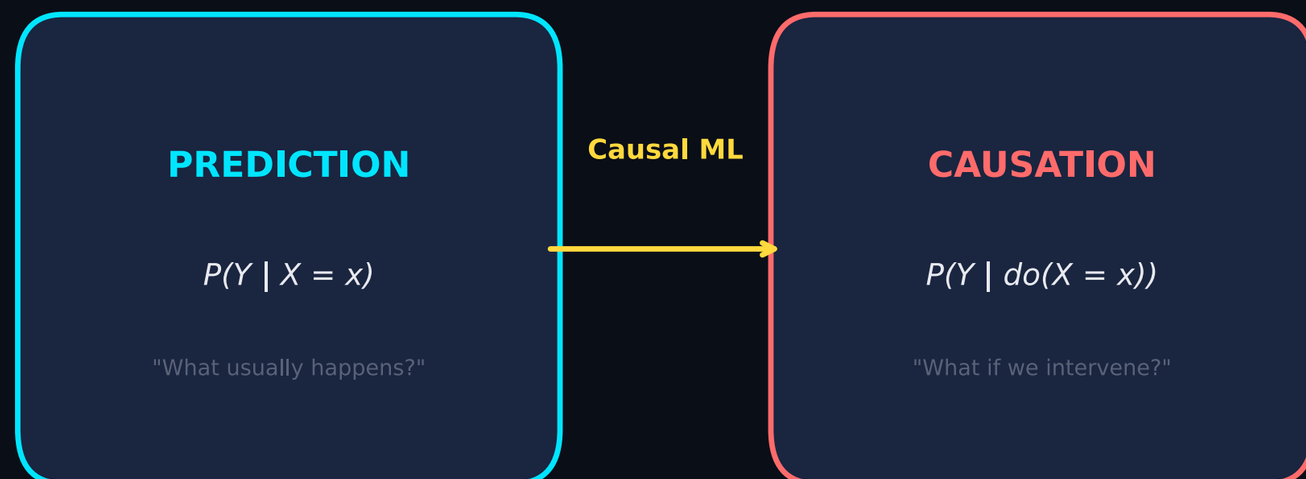
“The difficulty is to detach the framework of fact — of absolute fact — from the embellishments of theorists…“

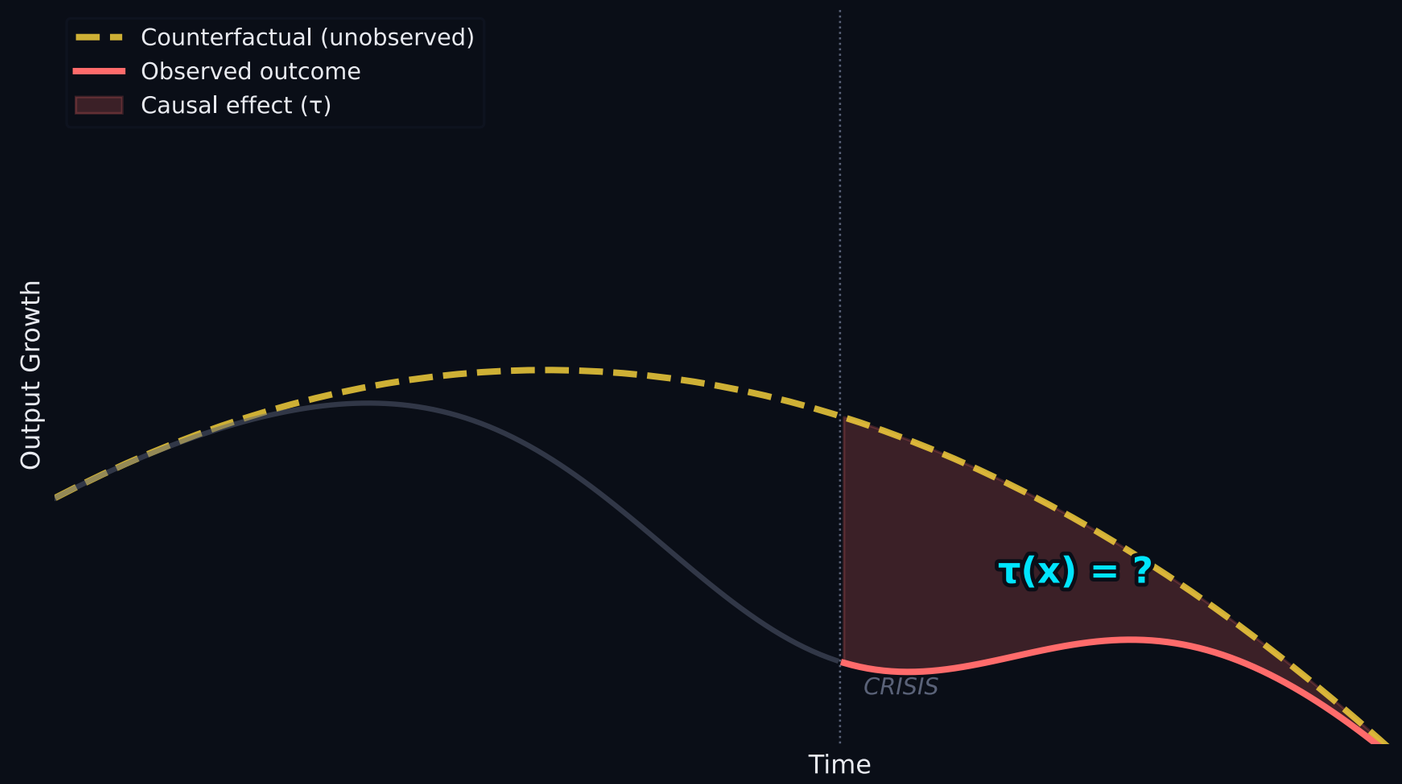
— The Memoirs of Sherlock Holmes

The Core Tension



A crowing rooster is a good **predictor** of sunrise. But intervening on the rooster won't stop the sun from rising.

We Never See the Path Not Taken



The **fundamental problem** of causal inference: we observe one outcome per unit.

Counterfactual predictions can never be validated — we need them to be **plausible**.

Why ML Changes the Game

CONFOUNDER SELECTION

LASSO selects which controls matter from hundreds of candidates. Double selection (Belloni et al., 2014) selects on **both** outcome and treatment.

COUNTERFACTUAL QUALITY

ML propensity scores build better matches between treated and control groups. Better E[Y|X] directly improves counterfactual estimates.

HETEROGENEOUS EFFECTS

Causal Forests find who benefits and who is harmed — nonparametrically. No need to pre-specify interactions or functional forms.

The Average Hides Everything

THE OLD QUESTION

“Does the drug work?”

One number for everyone. A single average treatment effect.

If the drug helps some patients and harms others, the average may be **zero** — and we’d conclude it doesn’t work.

THE NEW QUESTION

“For whom does it work, and why?”

A **personalised** effect for each patient — based on their characteristics.

We want to know: young patients benefit, elderly patients don’t. **That’s the signal.**

A **causal forest** is a machine that answers the new question.

How a Causal Forest Thinks

Imagine sorting patients into groups where the drug's effect is most **different**:

1. **Split** the data: find the characteristic that best separates *who benefits* from *who doesn't* — age? income? blood pressure? The algorithm searches, you don't have to guess.

2. **Keep splitting** to maximise the difference in treatment effects **between** groups. Each final leaf contains similar people — a mini-experiment with its own effect estimate.

3. **Repeat** hundreds of times with random subsets of data and variables. Average across all trees. The forest is **robust** where any single tree is fragile.

4. **Honesty trick**: one half of the data decides *where to split*; the other half *measures the effect*. This prevents overfitting and gives valid confidence intervals.

The Formal Idea

$$Y_i = \tau_{(i)} W_i + X_i \beta + \varepsilon_i$$

where $\tau_{(i)}$ is the treatment effect for unit i — and it **varies**

Standard regression: you choose the interactions

Causal forest: the algorithm finds them

KEY PROPERTIES

Splits maximise **heterogeneity of effects** between groups — not prediction accuracy.

Each leaf is a neighbourhood of **similar people**, yielding a local treatment effect with a valid confidence interval.

No functional form assumed.

APPLICATION

What is the cost of a **financial crisis?**

107 countries | 46 variables | 1985–2017

Risk vs. Vulnerability

RISK

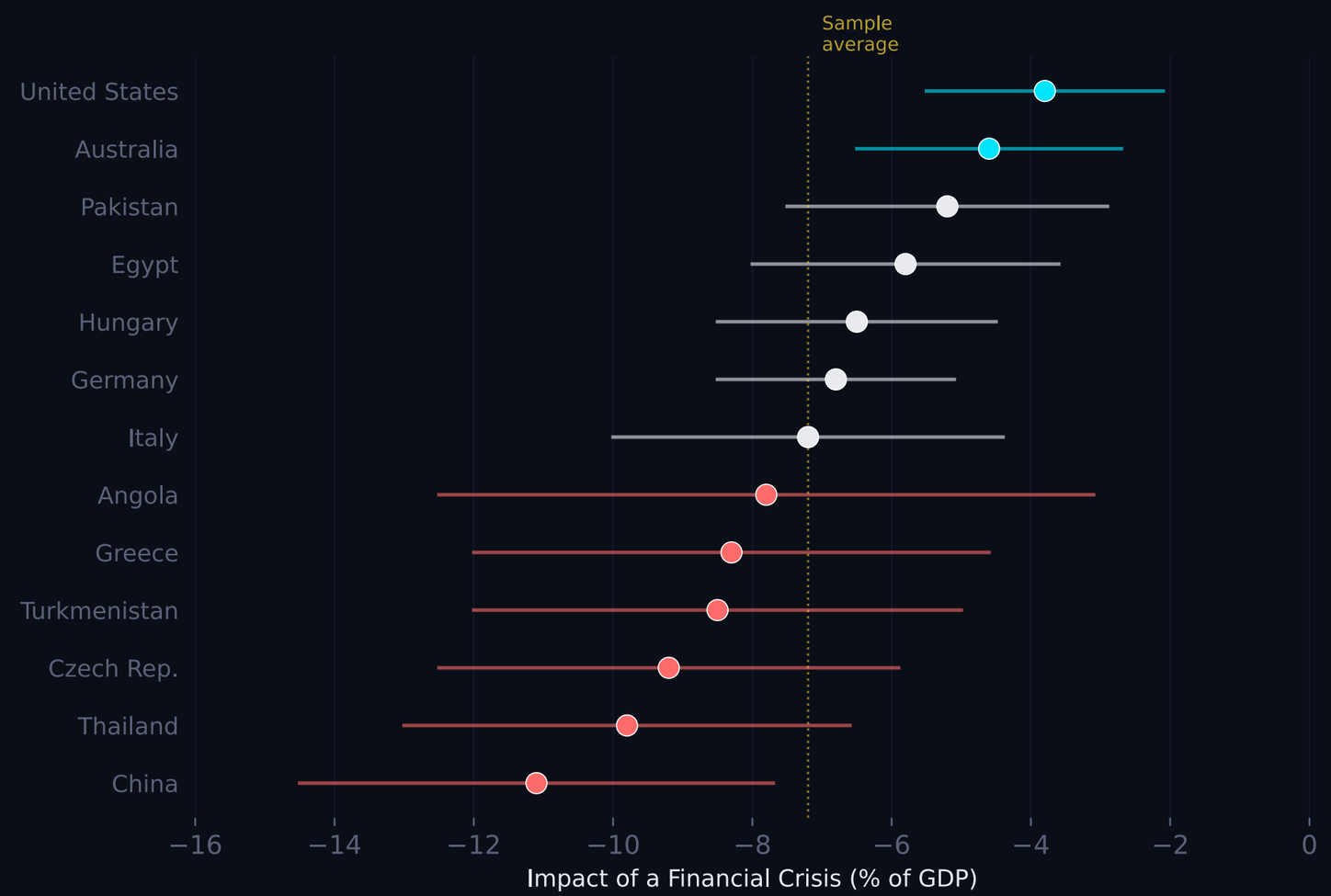
The probability that a crisis occurs. Most EWS models focus here. *Will the storm hit?*

VULNERABILITY

The cost of a crisis, **if** it occurs. This paper focuses here. *How sturdy is the house?*

Two houses on the Florida seaboard face the same **risk**. A concrete house is less **vulnerable** than a wooden one.

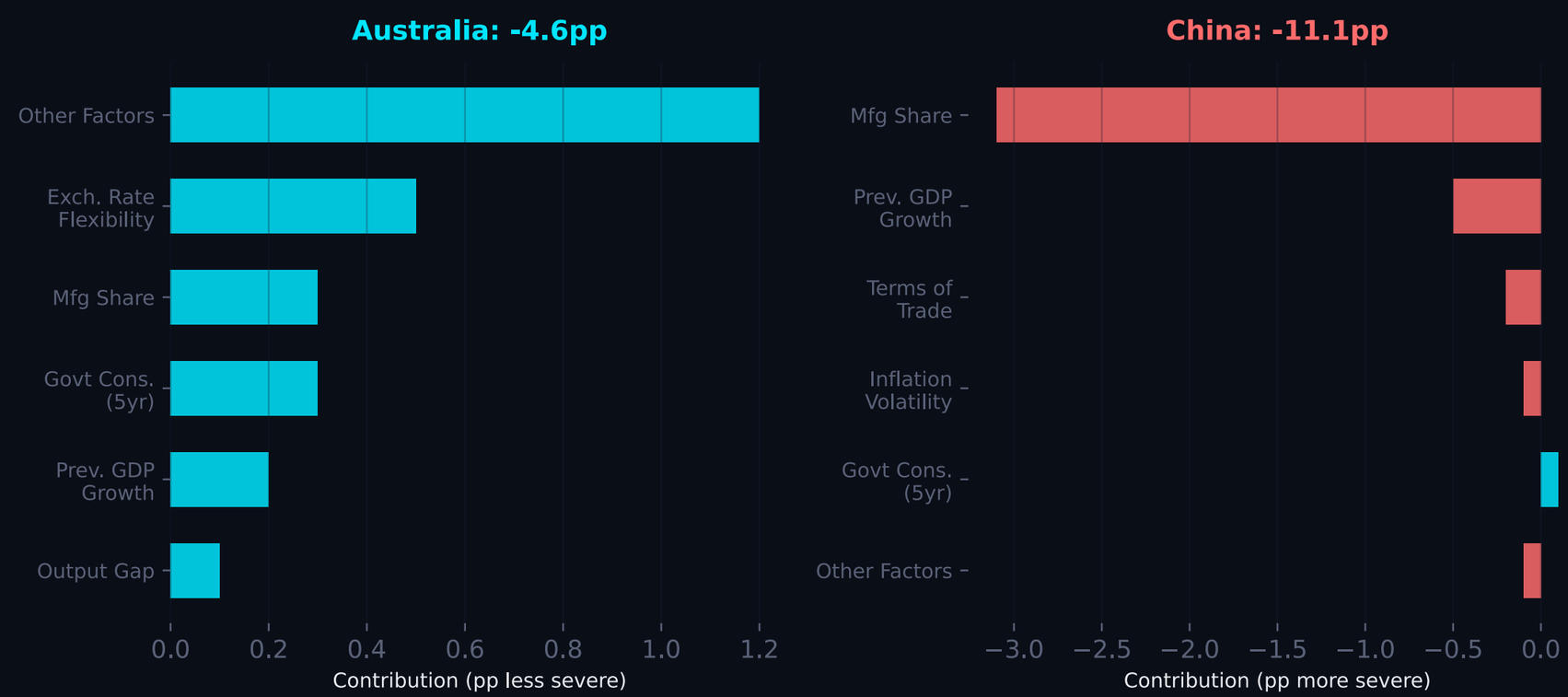
The Cost Varies Enormously



Sample average: **-7.2 pp** of GDP over two years

But ranges from **-3.8** (United States) to **-11.1** (China)

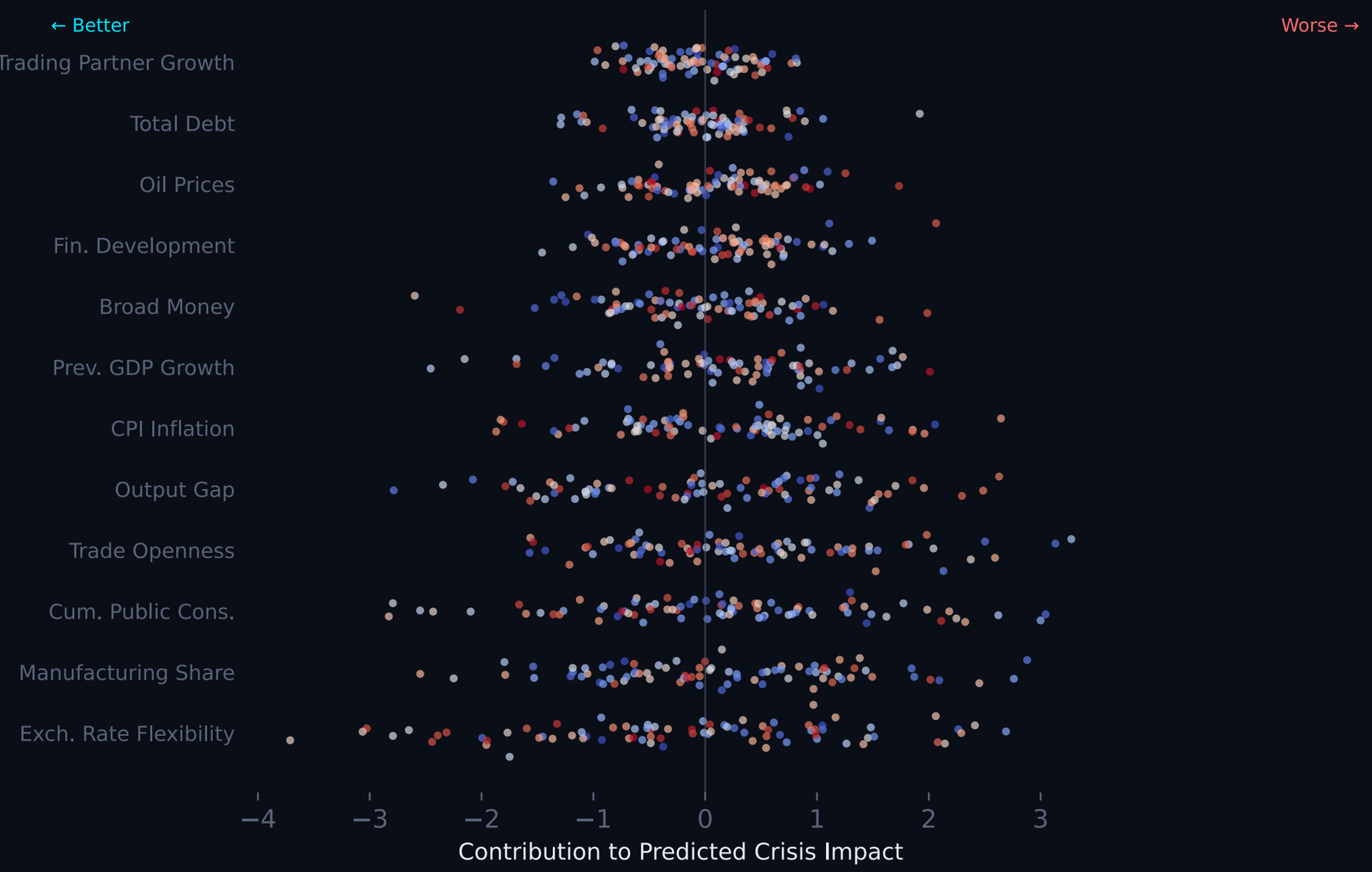
Decomposing the Black Box



Shapley values decompose each prediction into variable contributions

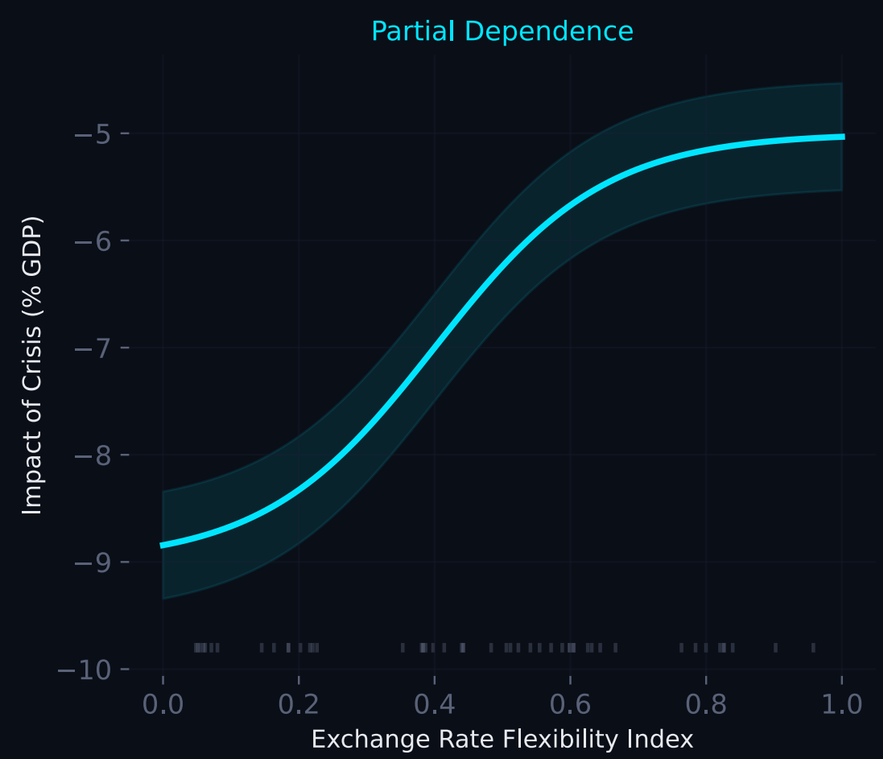
Australia: flexible exchange rate **buffers**. China: manufacturing share **amplifies**.

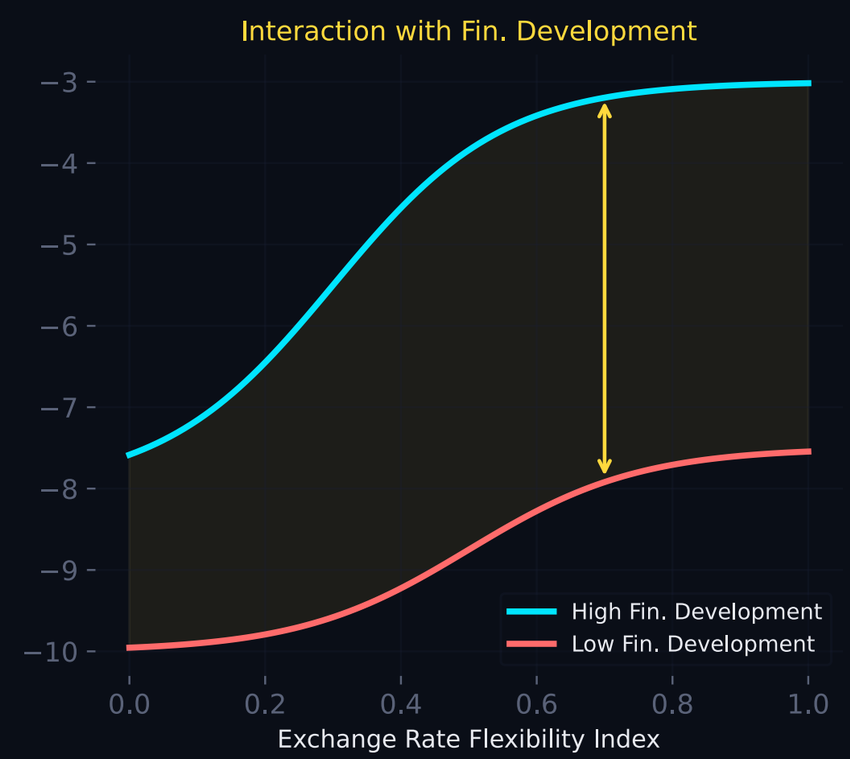
What Matters Most?



Exchange rate flexibility is the single most important moderator

The Exchange Rate Story





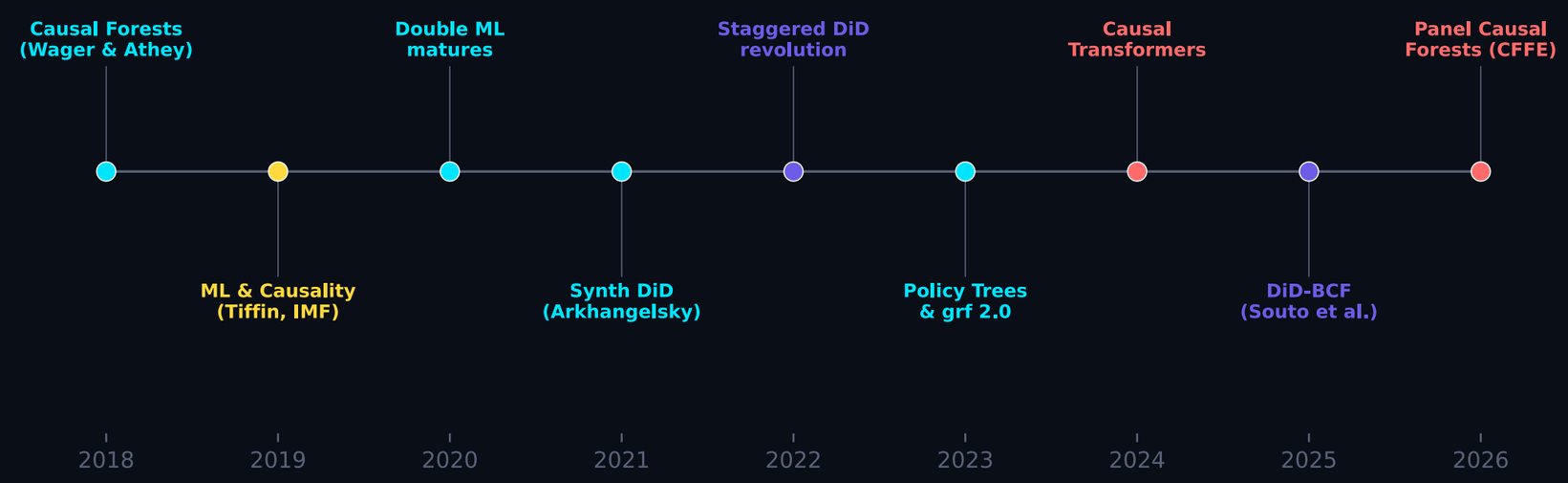
Flexibility reduces crisis cost — but **only for financially developed countries**

The interaction is **nonlinear** — undetectable by standard regression

SIX YEARS LATER

Where has the **field gone?**

The Causal ML Frontier



From estimating **average** effects to estimating **who is affected and why**

The New Toolkit

DOUBLE / DEBIASED ML — Chernozhukov et al. (2018)

ML for nuisance parameters + semiparametric efficiency. Clean ATEs in high dimensions.

PANEL CAUSAL FORESTS — CFFE (Aytug et al., 2026)

Node-level residualisation of fixed effects. HTEs from panel data with staggered treatment.

STAGGERED DiD + ML — DiD-BCF (Souto et al., 2025)

Bayesian causal forests with parallel trends. Full posterior on CATEs.

CAUSAL TRANSFORMERS — ICML 2026

Cross-attention for long-range confounders. Counterfactuals over time series.

What This Paper Would Look Like Today

2019 Approach	2026 Approach
Cross-sectional causal forest	**Panel causal forest (CFFE)** with unit + time FE
SMOTE for class imbalance	**Honest subsampling** with staggered treatment timing
Shapley values for interpretation	**SHAP + causal mediation** via DAG-structured decomposition
Single treatment (crisis dummy)	**Continuous treatment** (crisis severity) via instrumental forests
Post-hoc interaction exploration	**Policy trees** for optimal intervention assignment

The core insight — **heterogeneous vulnerability** — is more relevant than ever

The Macro Gap

Causal forests have been applied extensively in:

- Labor economics (job training RCTs, wage effects)
- Development economics (education outcomes)
- Health (treatment heterogeneity)

But **barely touched** in:

- Macroeconomic growth
- Trade and industrial policy
- Monetary and fiscal policy transmission
- Resource wealth and TFP

The 2019 crisis paper remains one of very few macro applications.

KEY TAKEAWAYS

1. Averages hide everything
2. Nonlinearities and interactions are the signal, not the noise
3. The frontier has moved — but macro hasn't caught up
4. Panel causal forests are ready for cross-country questions

"More has been learned about causal inference in the last few decades than the sum total of everything that had been learned about it in all prior recorded history."

— Gary King, 2007

The tools exist. The macro questions are waiting.